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INFORMS 2026, San Francisco

Decision-Aware Segmentation

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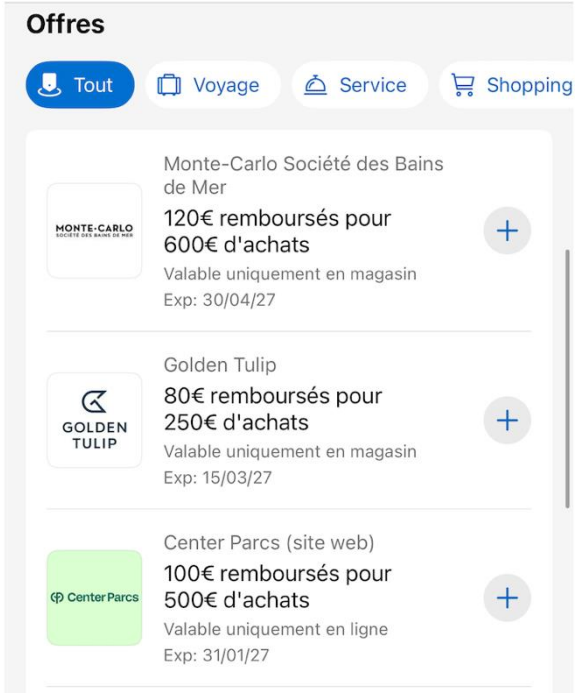
Customer Segmentation Drives Business Decisions

- Department stores, online retailers, banks (e.g., Sephora, Amazon, American Express): coupon campaigns, ad campaigns

SEPHORA

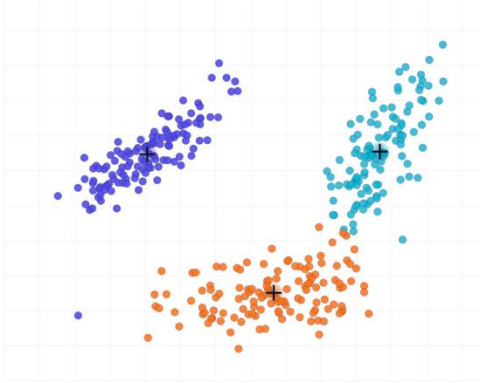
amazon

AMERICAN EXPRESS

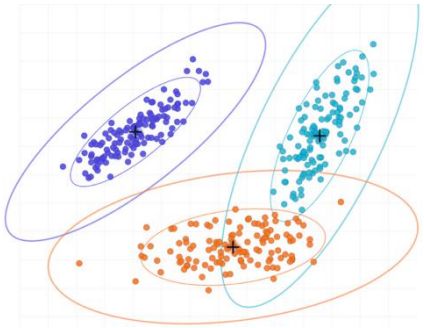


Widely-used segmentation algorithms optimize statistical fit

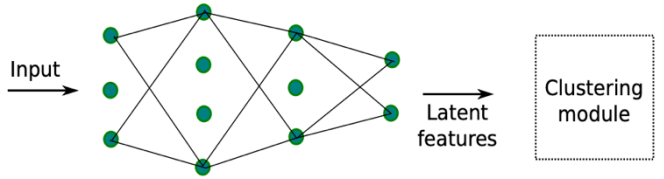
Centroid-based clustering
(e.g., K-means)



Model-based clustering
(e.g., GMM)



ML/DL-based clustering



Are we segmenting for the right objective?

Segmentation Should Optimize Decision Outcomes



- **Prescriptive segmentation** (Liu et al., 2010; Aouad et al., 2023; Zhang and Misra, 2024; Cui and Hamilton, 2026)

- **Uplift modeling** (Ascarza, 2018; Lemmens and Gupta, 2020; …)
- **Policy learning** (Manski, 2004; Tetenov, 2012; Kitagawa, 2018; Athey and Wager, 2021; …)
- **Decision-aware learning** (Mandi et al., 2024; Elmachtoub et al., 2020, 2022; Vanderschueren et al., 2022; …)

Running Example

- Suppose an e-commerce retailer wants to determine how to allocate Ad A and Ad B across customers to maximize conversion.

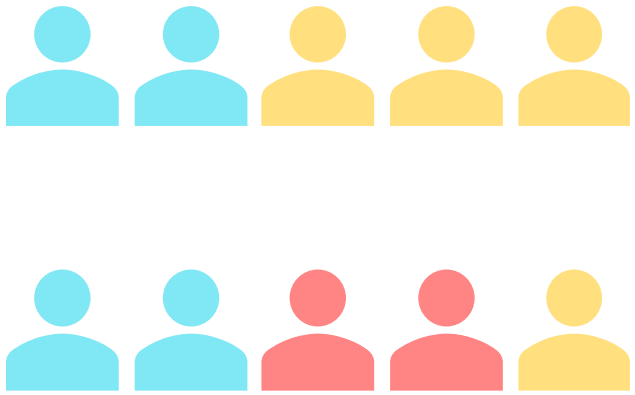
- The firm can choose to show Ad A or Ad B to each customer.

- Full personalization?
  - over-fitting due to insufficient data; costly to obtain data
  - difficult to operationalize

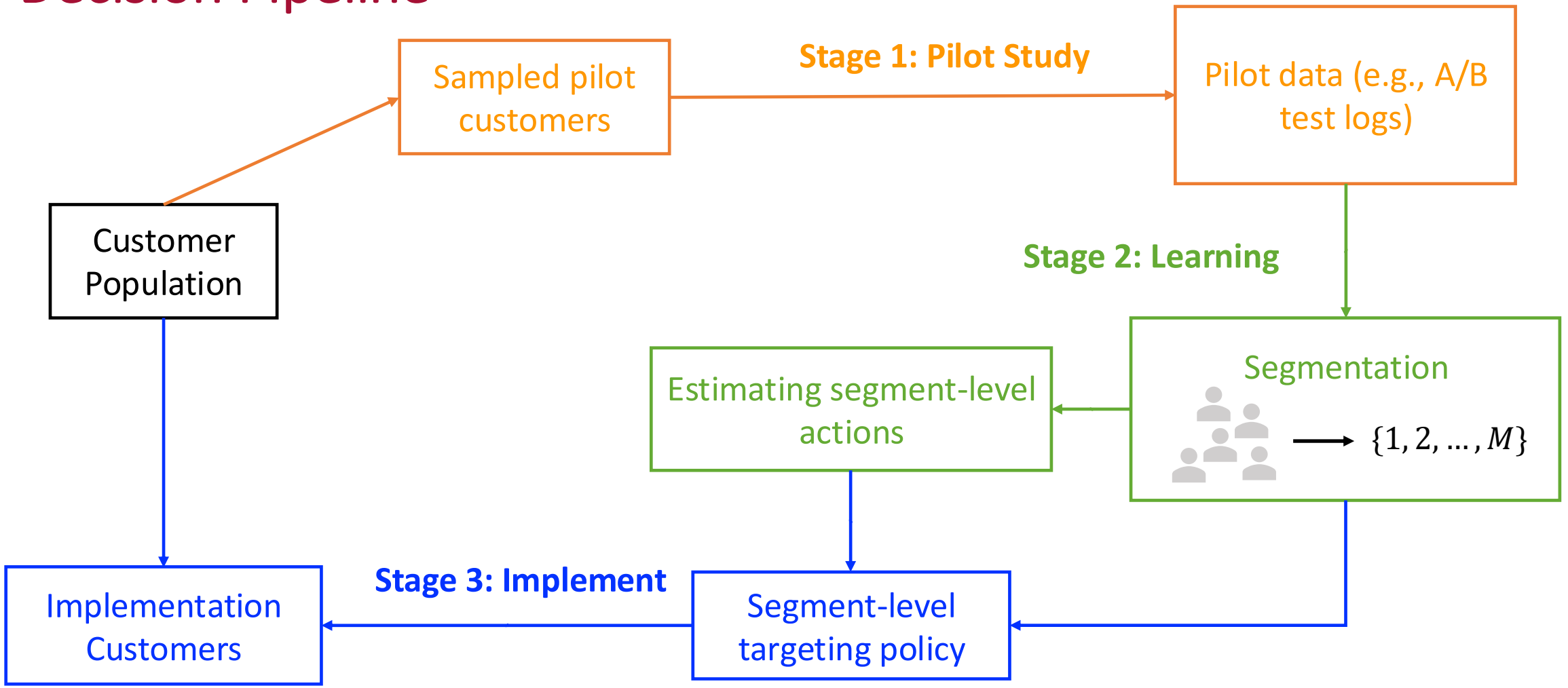
- The firm wants to deploy a **segment-level targeting** instead.

Research Questions

How should firms segment to optimize decision outcomes?

How many segments	How to segment
	

Decision Pipeline

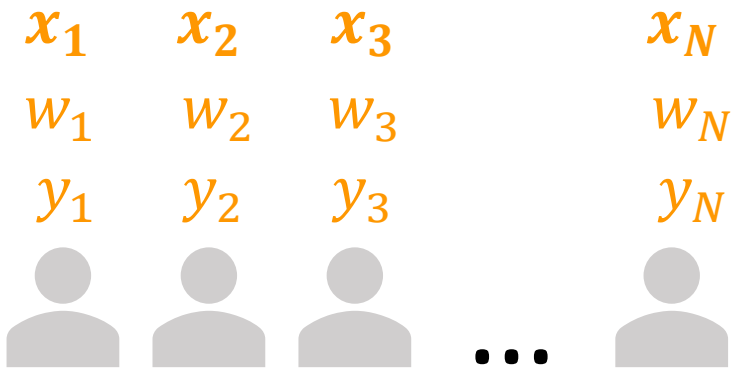


How to segment customers to maximize implementation outcomes?

Problem Setup

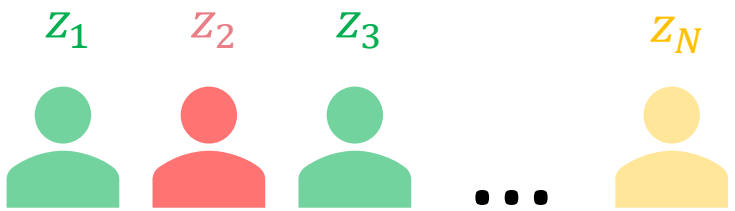
(Observables: pilot data)

- Each customer i with a feature vector $\boldsymbol{x}_i \in \mathbb{R}^d$
- Treatment $w_i \in \{0,1\}$, and observed conversion $y_i \in \{0,1\}$



(Unobservables: latent structure)

- K latent segments: each segment k characterized by (f_k, θ_k) .
- Each customer i belongs to a latent segment $z_i \in \{1, 2, .., K\}$



(Data generation process)

- The observed conversion y_i for customer i follows a logistic model:

$$y_i \sim \text{Bernoulli}\,(p_i),$$

observed by managers

$$p_i = \text{Sigmoid}(\, f_{z_i}(\boldsymbol{x}_i) + \theta_{z_i} \cdot w_i)$$

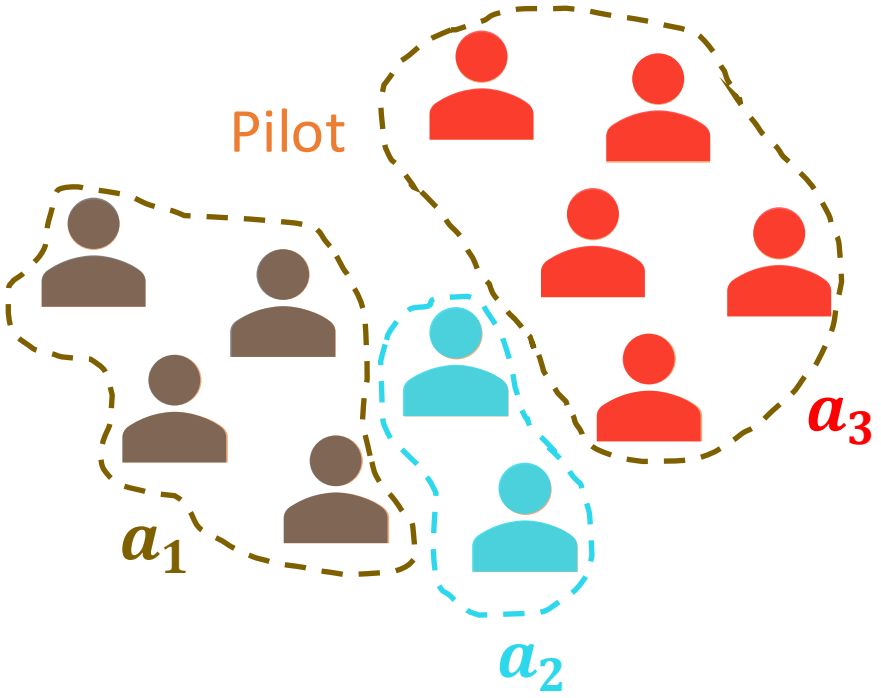
baseline

treatment parameter

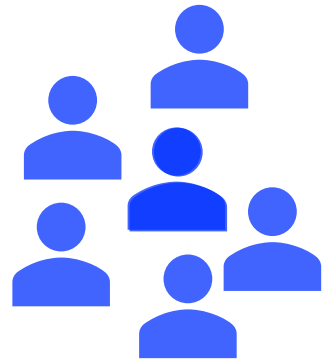
Not known by managers

Manager's Optimization Problem

The manager learns a segmentation ϕ from the pilot data $\{(x_i, w_i, y_i)\}$, and estimates targeting actions for each segment.



Implementation



$$r_j = \sigma\left(f_{z_j}(x_i) + \theta_{z_j} \cdot a_3 \right)$$

The manager seeks a segmentation ϕ that maximizes the sum of probabilities

$$\max_{\phi} \sum_{j \in \text{implement}} r_j$$

Optimization Proxy

The manager's optimization goal:

Not observable during the learning stage!

$$\max_{\text{segmentation } \phi} \sum_{j \in \text{Implement}} r_j \rightarrow \hat{r}_j$$

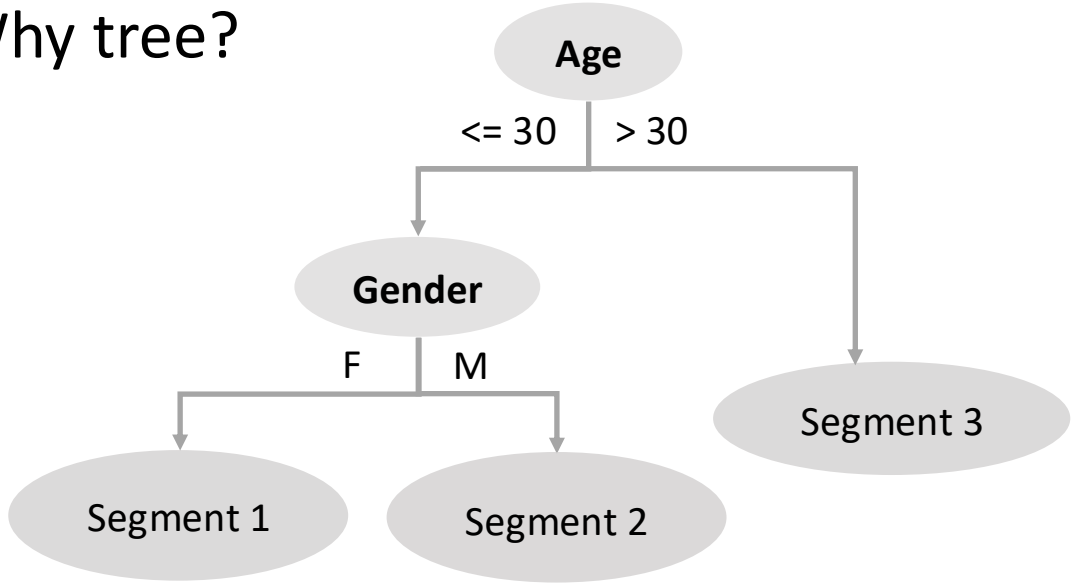
(customer j, action a_j) ⟶ doubly-robust (DR) estimator ⟶ $\hat{r}_j$

per-customer, per-action estimate of counterfactual outcomes (Dudík, M. et al., 2011).

Decision-Aware Segmentation (DAS)

A framework that produces tree-based segmentation by directly maximizing the downstream outcomes.

Why tree?



- Interpretable segmentation structure
- Non-linear relationships
- Easy to allocate

Decision-Aware Segmentation (DAS)

How many segments	How to segment
	Decision-Aware Segmentation Tree (DAST)

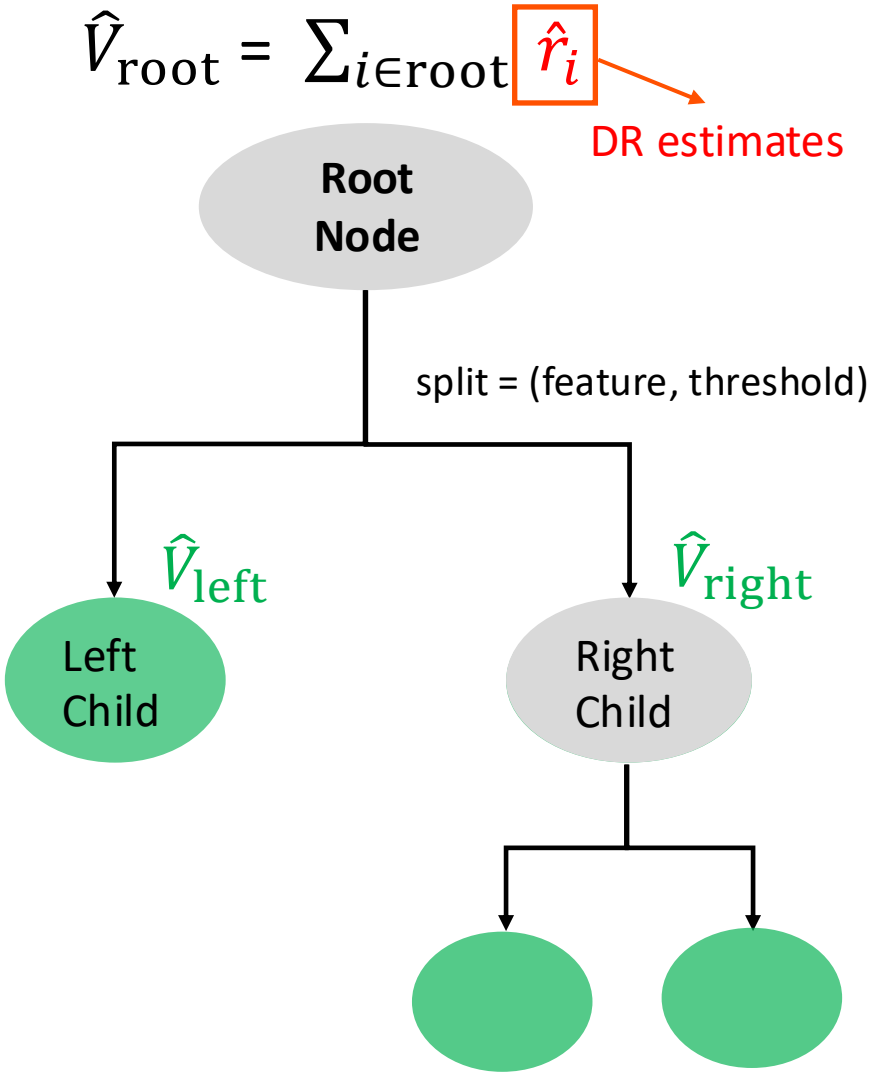
Decision-Aware Segmentation Tree

Build on MST (Aouad et al., 2023) and Policytree (Sverdrup et al., 2020),

- better aligned with managerial objectives
- more computationally efficient

How to grow a decision-aware segmentation tree
(simplified procedure for intuition)

- For each leaf, evaluate all split candidates by policy-value gain $\Delta V = \hat{V}_{\text{left}} + \hat{V}_{\text{right}} - \hat{V}_{\text{parent}}$
- Jointly choose the leaf node and the split that yields the largest gain $\Delta V_{\max}$
- Repeat the above steps until the tree grows to the desired number of leaves M



Decision-Aware Segmentation (DAS)

How many segments	How to segment
Decision-Aware Model Selection	Decision-Aware Segmentation Tree

Decision-Aware Model Selection

Agnostic to the segmentation algorithm

How to choose the right number of segments

(Simplified procedure for intuition)

1. Specify candidate segment counts, e.g., $\mathcal{M} = \{1, 2, 3, .., 10\}$

2. For each $M \in \mathcal{M}$,

   - learn a segmentation ϕ_M (DAST)
   - calculate $\hat{V}(\phi_M)$ on a held-out validation set: $\hat{V}(\phi_M) = \sum_{j \in \text{validation}} \hat{r}_j$ ($\hat{r}_j$: DR estimates)

3. Select the M^* with the highest value $M^* \in \underset{M \in \mathcal{M}}{\text{argmax}}\, \hat{V}(\phi_M)$

Theoretical Guarantees

Can DAST **scale** to millions of customers?

In math language: Time complexity of DAST: $O(d \cdot N \log N + d^2 N)$

Make it sound friendlier: DAST scales efficiently

- log-linear in the size of the training set
- quadratic in the covariate dimensionality

Theoretical Guarantees

Does DAST choose the **right** split?

In math language:

$$\text{Regret} = V(h^*) - V(\hat{h}) \leq C \cdot \sqrt{\log\left(\frac{16 \cdot |\mathcal{H}|}{\delta}\right)}$$

Make it sound friendlier:

With high probability, the split chosen by DAST achieves a targeting performance close to the best possible candidate under perfect information

Theoretical Guarantees

How many validation **samples** do we need to **reliably choose** the number of segments?

In math language:

$$N \geq C \cdot \frac{1}{\eta^2} \log\left(\frac{2 \cdot |\mathcal{M}|}{\delta}\right)$$

In English

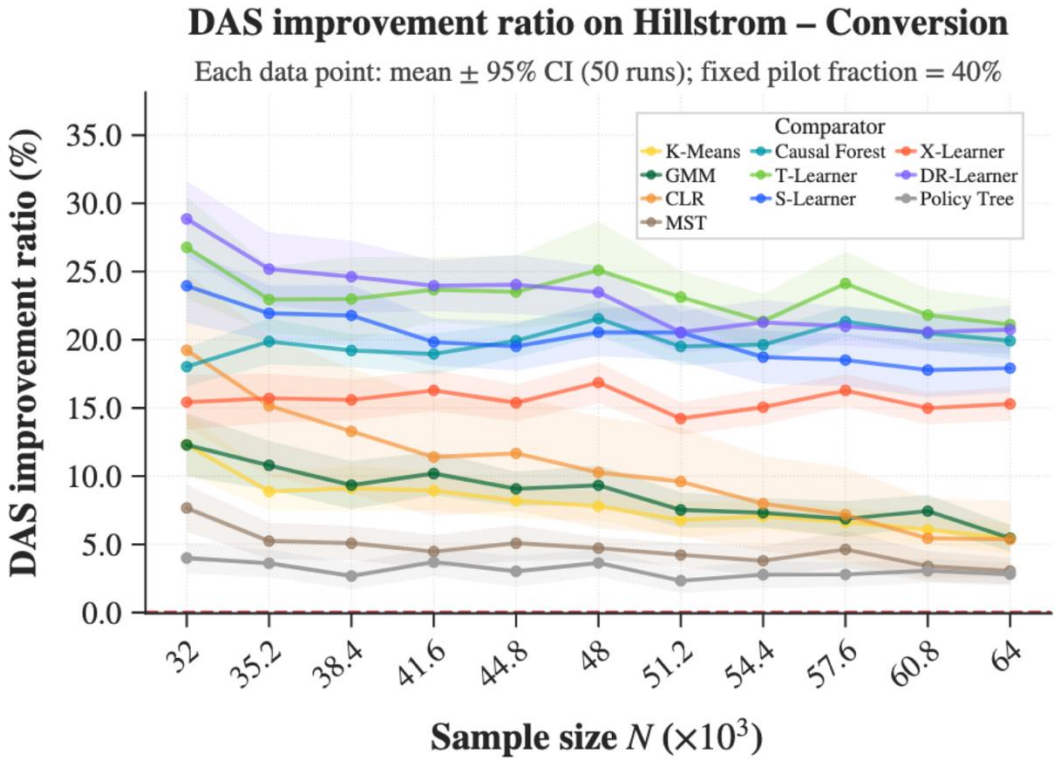
- The required validation sample size grows **logarithmically**
- Algorithm agnostic

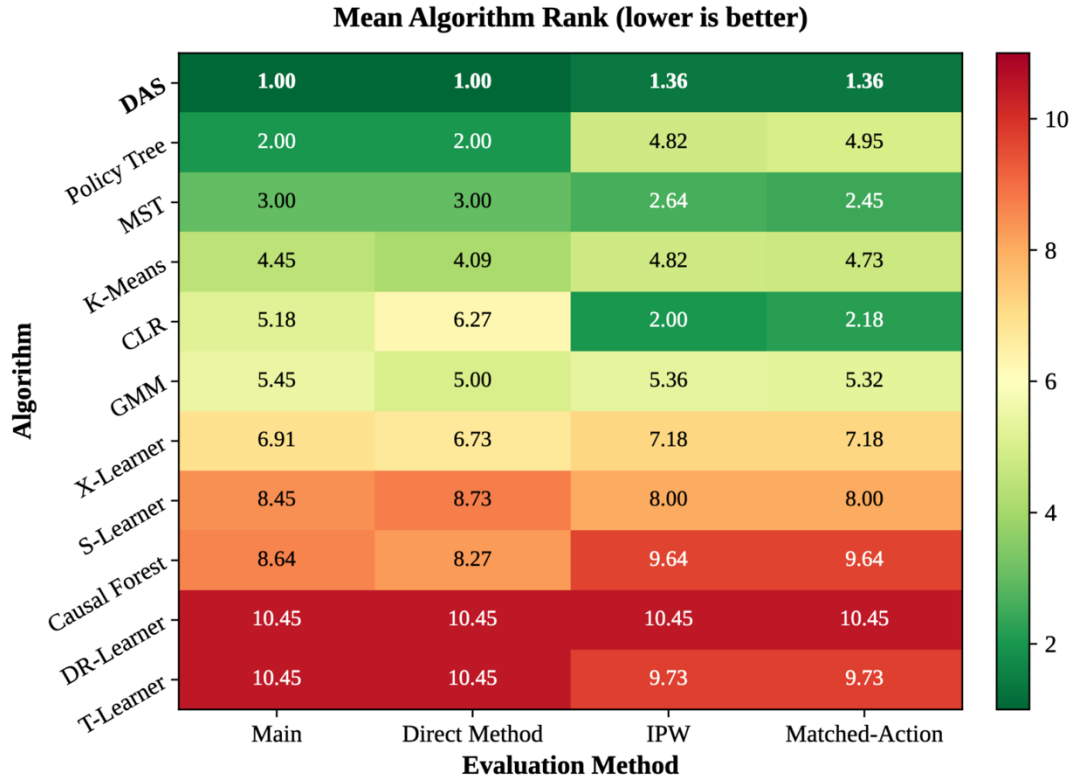
Numerical Studies: Benchmark Methods

Segmentation Methods	HTE-based Methods
➢ **K-Means** (Hartigan and Wong, 1979) ➢ **GMM** (Dempster et al., 1977) ➢ **Clusterwise-Linear Regression** (DeSarbo and Cron, 1988) ➢ **Market Segmentation Trees** (Aouad et al., 2023)	➢ **Meta-learners**, including **T-learner, S-learner, X-learner, and DR-learner** (Künzel et al., 2019; Kennedy, 2023) ➢ **Causal Forest** (Wager and Athey, 2018) ➢ **Policytree** (Sverdrup et al., 2020)

Dataset: Criteo online advertising A/B tests (RCT)

- **N**: 140,000 customers
- **X**: 12 features
- **Treatment**: binary (0/1)
- **Y**: conversion rate (~0.3%, very sparse outcomes)





Open Source (Python + R Packages coming soon!)

- One-step installation: `pip install XXX` or `install.packages("XXX")`
- Minimal setup: 4 lines of code

```
from das import DASTree

tree = DASTree(M=3)
tree.fit(X, y, D)

actions  = tree.predict_action(X_new)
```

→ Prescriptive actions

- Support multi-action settings
- Easy integration with existing targeting pipelines

Takeaways: Optimizing segmentation for the decisions

- **A new perspective:** Segmentation should be optimized for decisions
- **A new solution:** DAS framework
- **A deployable tool:** Open-source Python/R packages

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Thank You & Questions

My research interests:

- **AI for Decisions**: Operations & ML
- **AI for Science**: Human–AI collaboration

Open to discussions and collaboration: ke.li@insead.edu